RCS Day 5

Switch-controlled lighting, de-energized troubleshooting, correction, and retest

FIND THE FIRST FAILED CHECKPOINT.

One expected path. One controlled fault. One correction at a time.

Today the goal is not random fixes. The goal is evidence that explains the failure.

Today's flow

Quiz -> model -> guided practice -> capstone lab -> QC



Morning: trace line, switch leg, load, neutral return, and grounding.
Afternoon: add switch/light boxes and diagnose an inserted fault.
End: correction, retest, evidence, label, reset.

Today's targets

What successful work must demonstrate

Line source path

Switch control

Switch leg to load

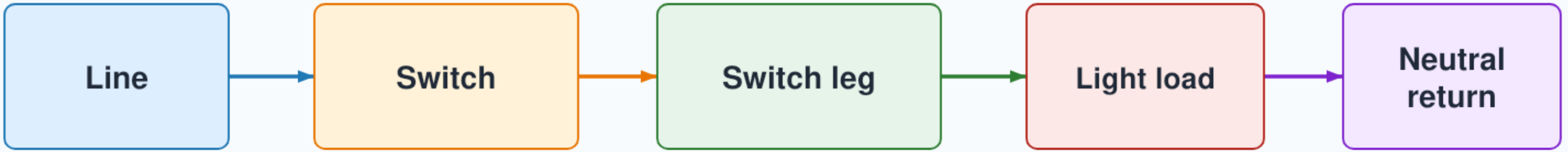
Light load

Neutral + ground paths

Trace a de-energized switch-controlled lighting path
Install a training switch and light outlet box
Use visual inspection and continuity to diagnose inserted faults
Document the fault, evidence, correction, and retest

The day's path

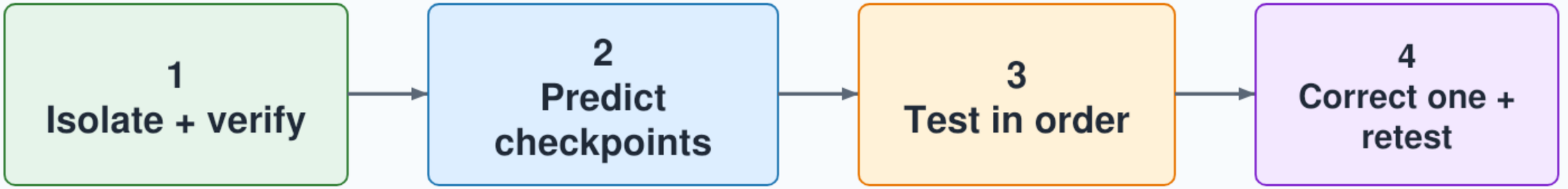
Trace the work before building it



Trace the path before testing. The first checkpoint that stops matching expectation narrows the fault location.

Controlled sequence

Release each stage only after its check



Students perform no energized diagnostics. Continuity is used only on isolated conductors under instructor control.

Working roles

6 students; every result receives another set of eyes

Builder
installs switch + light

Inspector
checks path + splices

Documenter
logs tests + correction

Rotate before the faulted state is issued.
The troubleshooter says the expected result before each test.
No one changes two conditions at once.

Verification checkpoints

Do not trust appearance alone

Expected path

Known source state

First mismatch

Correction + retest

A continuity beep is evidence of a path, not permission to energize.
Record probe points, expected result, observed result, and conclusion.

Safety boundaries

The lab begins only inside these controls

Mock-up isolated

No energized diagnostics

Continuity only isolated

Instructor controls fault

Treat any unverified conductor as energized. Replace damaged devices or conductors. Close and label the training system after approval.

Professional judgment

Safe work pauses when the evidence conflicts

Good method
predict -> test -> locate -> correct

Bad method
change everything -> lose evidence

Troubleshooting is controlled comparison. One change at a time preserves cause-and-effect evidence.

Quality criteria

Good physical work leaves visible proof

Trace line-switch-load-neutral first.
Prepare splices and pigtails cleanly.
Inspect boxes and terminations.
Test only the isolated expected path.
Correct, retest, document, and obtain approval.

No expected result? Do not place the probes.

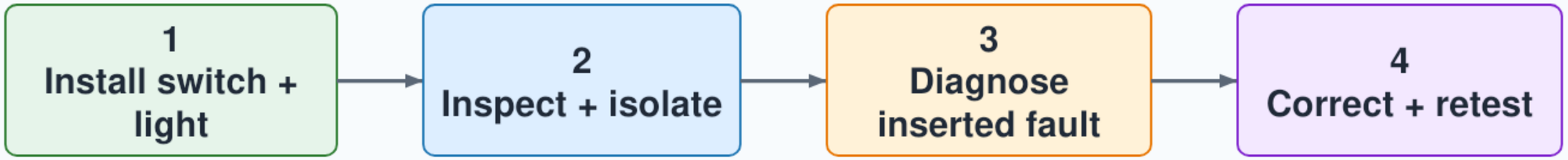
Lab release checklist

The station earns work only after the checks

Energy state verified and called out.
Switch/light path matches the diagram.
Boxes, cable, and splices inspected.
Expected checkpoints written before testing.
Continuity used only while isolated.
Instructor controls fault insertion and reset.

Afternoon lab

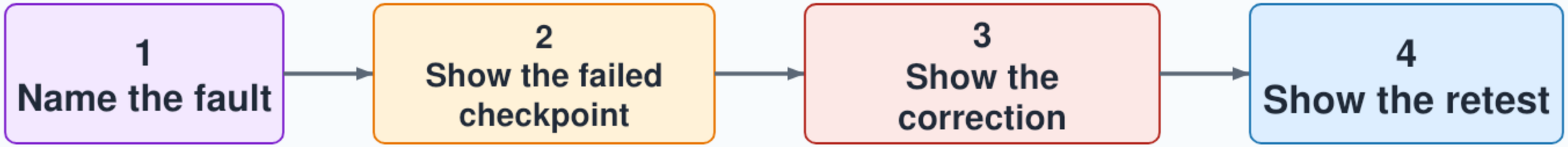
Slow sequence. Clear callouts. No surprises.



Add the switch and light point to the capstone, then diagnose one instructor-inserted open or misconnection.

Closeout order

End verified, documented, and reset



The repair is not complete until the result is retested, documented, labeled, and instructor-approved.

Evidence log

What students write down

Energy	isolated + known	Initials: _____
Path	line to neutral	Pass: _____
Visual	boxes + splices	Pass: _____
Checkpoint	first mismatch	Point: _____
Correction	one change	Action: _____
Retest	expected restored	Result: _____

A guess is not a diagnosis. Record the checkpoint that proves the fault and the retest that proves the correction.

Exit ticket

Three checks before dismissal

1. Draw line -> switch -> load -> neutral return.

2. Identify the first failed checkpoint in your fault.

3. State the evidence that proved your correction.

System isolated. Fault logged. Correction retested. Boxes closed.